

--- Tuning inc model ---

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| Iter | Eval   | Objective   | Objective   | BestSoFar   | BestSoFar   | ✓
NumLearningC-| MinLeafSize | MaxNumSplits |
|       | result  |             | runtime     | (observed)  | (estim.)    | cycles ✓
|       |         |             |             |             |             |
=====
=====|
|   1 | Best   | 0.36773 | 3.0663 | 0.36773 | 0.36773 | ✓
92 |      155 |      2 |
|   2 | Best   | 0.36193 | 0.56043 | 0.36193 | 0.36263 | ✓
15 |      1 |      4 |
|   3 | Best   | 0.2487 | 4.3936 | 0.2487 | 0.24871 | ✓
118 |      3 | 4115 |
|   4 | Accept | 0.36792 | 0.44661 | 0.2487 | 0.24871 | ✓
17 |     637 |     63 |
|   5 | Accept | 0.3544 | 0.64746 | 0.2487 | 0.24871 | ✓
21 |     300 |    5139 |
|   6 | Accept | 0.36444 | 1.5299 | 0.2487 | 0.24871 | ✓
76 |      3 |      2 |
|   7 | Accept | 0.26435 | 0.64432 | 0.2487 | 0.24872 | ✓
17 |      1 |    5059 |
|   8 | Accept | 0.2742 | 0.43734 | 0.2487 | 0.24873 | ✓
12 |      8 |    5140 |
|   9 | Best   | 0.24696 | 17.516 | 0.24696 | 0.24703 | ✓
476 |      2 |    4503 |
|  10 | Accept | 0.27324 | 0.56232 | 0.24696 | 0.24693 | ✓
12 |      2 |    1562 |
|  11 | Accept | 0.25063 | 19.701 | 0.24696 | 0.24696 | ✓
497 |      4 |    5081 |
|  12 | Accept | 0.24986 | 7.172 | 0.24696 | 0.24746 | ✓
171 |      2 |    5058 |
|  13 | Best   | 0.24638 | 15.022 | 0.24638 | 0.2466 | ✓
388 |      2 |    4850 |
|  14 | Best   | 0.24522 | 16.665 | 0.24522 | 0.24547 | ✓
456 |      3 |    3478 |
|  15 | Accept | 0.29198 | 0.38328 | 0.24522 | 0.2454 | ✓
11 |     17 |      709 |
|  16 | Accept | 0.24599 | 14.03 | 0.24522 | 0.24551 | ✓
401 |      3 |    3438 |
|  17 | Accept | 0.24966 | 16.367 | 0.24522 | 0.24724 | ✓
434 |      3 |    3833 |
|  18 | Accept | 0.24638 | 17.358 | 0.24522 | 0.24698 | ✓
496 |      2 |    2098 |
|  19 | Best   | 0.24309 | 16.214 | 0.24309 | 0.2452 | ✓
459 |      2 |    2646 |
|  20 | Best   | 0.24251 | 17.595 | 0.24251 | 0.24466 | ✓
490 |      1 |    2697 |

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| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | 
NumLearningC-| MinLeafSize | MaxNumSplits | 
| result | runtime | (observed) | (estim.) | cycles 
=====
| 21 | Accept | 0.2456 | 18.081 | 0.24251 | 0.24479 | 
499 | 1 | 3005 | 
| 22 | Accept | 0.24966 | 16.092 | 0.24251 | 0.24536 | 
448 | 1 | 2383 | 
| 23 | Accept | 0.24676 | 17.792 | 0.24251 | 0.24528 | 
496 | 1 | 4901 | 
| 24 | Accept | 0.39092 | 0.20819 | 0.24251 | 0.24534 | 
10 | 2569 | 1 | 
| 25 | Accept | 0.39092 | 0.19737 | 0.24251 | 0.24532 | 
10 | 2482 | 4363 | 
| 26 | Accept | 0.38512 | 0.22593 | 0.24251 | 0.24536 | 
10 | 47 | 1 | 
| 27 | Accept | 0.34454 | 0.3114 | 0.24251 | 0.24545 | 
10 | 32 | 19 | 
| 28 | Accept | 0.39092 | 0.21688 | 0.24251 | 0.24545 | 
10 | 2541 | 18 | 
| 29 | Accept | 0.38338 | 0.22241 | 0.24251 | 0.24548 | 
10 | 1 | 1 | 
| 30 | Accept | 0.39092 | 0.20014 | 0.24251 | 0.24552 | 
10 | 2585 | 438 | 

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Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 228.4465 seconds

Total objective function evaluation time: 223.8572

Best observed feasible point:

NumLearningCycles	MinLeafSize	MaxNumSplits
490	1	2697

Observed objective function value = 0.24251

Estimated objective function value = 0.24613

Function evaluation time = 17.5947

Best estimated feasible point (according to models):

NumLearningCycles	MinLeafSize	MaxNumSplits
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459

2

2646

Estimated objective function value = 0.24552

Estimated function evaluation time = 16.4587

--- Tuning damp model ---

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| Iter | Eval   | Objective   | Objective   | BestSoFar   | BestSoFar   |
NumLearningC-| MinLeafSize | MaxNumSplits |
|      | result  |             | runtime     | (observed)  | (estim.)    | cycles
|      |         |             |             |             |             |
=====
| 1 | Best   | 0.37894 | 1.9951 | 0.37894 | 0.37894 |
92 |      155 | 2 |
| 2 | Best   | 0.3629 | 0.37962 | 0.3629 | 0.36384 |
15 |      1 | 4 |
| 3 | Best   | 0.2572 | 4.22 | 0.2572 | 0.2572 |
118 |      3 | 4115 |
| 4 | Accept | 0.36483 | 0.39069 | 0.2572 | 0.25721 |
17 |      637 | 63 |
| 5 | Accept | 0.36676 | 0.34314 | 0.2572 | 0.25721 |
15 |      612 | 1740 |
| 6 | Accept | 0.27652 | 0.55999 | 0.2572 | 0.25722 |
14 |      4 | 3210 |
| 7 | Accept | 0.26957 | 0.64386 | 0.2572 | 0.25723 |
16 |      1 | 5107 |
| 8 | Best   | 0.25217 | 17.882 | 0.25217 | 0.25217 |
486 |      1 | 5135 |
| 9 | Accept | 0.25662 | 2.7384 | 0.25217 | 0.25216 |
72 |      1 | 929 |
| 10 | Best   | 0.25101 | 12.373 | 0.25101 | 0.25097 |
342 |      1 | 1525 |
| 11 | Accept | 0.25546 | 14.558 | 0.25101 | 0.25108 |
422 |      2 | 2051 |
| 12 | Accept | 0.25256 | 5.0888 | 0.25101 | 0.25149 |
143 |      1 | 2928 |
| 13 | Accept | 0.25449 | 8.2349 | 0.25101 | 0.25204 |
232 |      1 | 2363 |
| 14 | Accept | 0.25256 | 10.188 | 0.25101 | 0.25208 |
287 |      1 | 2641 |
| 15 | Accept | 0.39053 | 0.1994 | 0.25101 | 0.25205 |
10 |      2459 | 1 |
| 16 | Accept | 0.32715 | 0.2861 | 0.25101 | 0.25203 |
10 |      13 | 40 |
| 17 | Accept | 0.38744 | 0.24405 | 0.25101 | 0.252 |
10 |      21 | 1 |

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| 18 | Accept | 0.39053 | 0.19601 | 0.25101 | 0.25198 | ✓
10 | 2450 | 5160 |
| 19 | Accept | 0.30686 | 0.35475 | 0.25101 | 0.25188 | ✓
10 | 1 | 104 |
| 20 | Accept | 0.33256 | 0.31732 | 0.25101 | 0.25188 | ✓
10 | 78 | 5142 |
|===== ✓
=====|
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | ✓
NumLearningC-| MinLeafSize | MaxNumSplits |
| | result | | runtime | (observed) | (estim.) | cycles ✓
| | | | | | |
|===== ✓
=====|
| 21 | Accept | 0.25353 | 7.1618 | 0.25101 | 0.25216 | ✓
202 | 1 | 2911 |
| 22 | Accept | 0.39053 | 0.18808 | 0.25101 | 0.2522 | ✓
10 | 2548 | 180 |
| 23 | Accept | 0.31807 | 0.33032 | 0.25101 | 0.25221 | ✓
10 | 33 | 426 |
| 24 | Accept | 0.36213 | 0.25974 | 0.25101 | 0.2522 | ✓
10 | 262 | 5 |
| 25 | Accept | 0.39053 | 0.99857 | 0.25101 | 0.25212 | ✓
63 | 2570 | 5149 |
| 26 | Accept | 0.38667 | 0.21803 | 0.25101 | 0.2521 | ✓
10 | 1 | 1 |
| 27 | Accept | 0.39053 | 0.20049 | 0.25101 | 0.25211 | ✓
10 | 2534 | 11 |
| 28 | Accept | 0.28792 | 0.36823 | 0.25101 | 0.25211 | ✓
10 | 4 | 357 |
| 29 | Accept | 0.38512 | 1.1932 | 0.25101 | 0.25204 | ✓
65 | 1 | 1 |
| 30 | Accept | 0.27749 | 0.40247 | 0.25101 | 0.25205 | ✓
10 | 1 | 1032 |

```

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 96.94 seconds

Total objective function evaluation time: 92.5147

Best observed feasible point:

NumLearningCycles	MinLeafSize	MaxNumSplits
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342

1

1525

Observed objective function value = 0.25101

Estimated objective function value = 0.25284
 Function evaluation time = 12.3732

Best estimated feasible point (according to models):

NumLearningCycles MinLeafSize MaxNumSplits

287

1

2641

Estimated objective function value = 0.25205
 Estimated function evaluation time = 10.3086

--- Tuning freq model ---

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|=====
| Iter | Eval   | Objective   | Objective   | BestSoFar   | BestSoFar   |
NumLearningC-| MinLeafSize | MaxNumSplits |
|       | result  |             | runtime     | (observed)  | (estim.)    | cycles
|=====
=====
| 1 | Best | 0.20444 | 2.0844 | 0.20444 | 0.20444 |
92 | 155 | 2 |
| 2 | Best | 0.19111 | 0.43112 | 0.19111 | 0.19189 |
15 | 1 | 4 |
| 3 | Best | 0.084831 | 3.6802 | 0.084831 | 0.084838 |
118 | 3 | 4115 |
| 4 | Accept | 0.21295 | 0.38592 | 0.084831 | 0.084841 |
17 | 637 | 63 |
| 5 | Accept | 0.19266 | 0.54394 | 0.084831 | 0.084841 |
24 | 496 | 1883 |
| 6 | Accept | 0.25488 | 1.1496 | 0.084831 | 0.08484 |
62 | 4 | 1 |
| 7 | Best | 0.083285 | 4.7804 | 0.083285 | 0.083286 |
166 | 2 | 5039 |
| 8 | Accept | 0.094686 | 0.38985 | 0.083285 | 0.083272 |
12 | 1 | 1001 |
| 9 | Accept | 0.099324 | 0.37928 | 0.083285 | 0.083264 |
12 | 9 | 4910 |
| 10 | Accept | 0.090435 | 0.47315 | 0.083285 | 0.083413 |
13 | 1 | 4531 |
| 11 | Accept | 0.093527 | 0.4006 | 0.083285 | 0.083254 |
11 | 2 | 5130 |
| 12 | Best | 0.08 | 17.329 | 0.08 | 0.079978 |
465 | 1 | 2759 |
| 13 | Accept | 0.089855 | 0.66951 | 0.08 | 0.079972 |
20 | 8 | 476 |
| 14 | Accept | 0.082705 | 6.9917 | 0.08 | 0.079984 |
222 | 4 | 984 |

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| 15 | Accept | 0.087536 | 12.209 | 0.08 | 0.079981 | ✓
445 | 10 | 1152 |
| 16 | Accept | 0.10261 | 0.31603 | 0.08 | 0.079978 | ✓
10 | 3 | 135 |
| 17 | Accept | 0.080966 | 12.949 | 0.08 | 0.080103 | ✓
416 | 1 | 4883 |
| 18 | Accept | 0.080386 | 14.803 | 0.08 | 0.080106 | ✓
473 | 2 | 2659 |
| 19 | Best | 0.079614 | 13.969 | 0.079614 | 0.079906 | ✓
477 | 2 | 2668 |
| 20 | Accept | 0.081159 | 14.463 | 0.079614 | 0.080213 | ✓
495 | 2 | 3347 |
|===== ✓
=====|
| Iter | Eval | Objective | Objective | BestSoFar | BestSoFar | ✓
NumLearningC-| MinLeafSize | MaxNumSplits |
| result | runtime | (observed) | (estim.) | cycles ✓
| | |
|===== ✓
=====|
| 21 | Accept | 0.08058 | 16.478 | 0.079614 | 0.080256 | ✓
486 | 2 | 1671 |
| 22 | Accept | 0.080386 | 14.056 | 0.079614 | 0.080257 | ✓
414 | 1 | 2705 |
| 23 | Best | 0.078841 | 15.44 | 0.078841 | 0.079727 | ✓
470 | 1 | 2197 |
| 24 | Best | 0.078454 | 15.1 | 0.078454 | 0.078806 | ✓
492 | 1 | 1617 |
| 25 | Accept | 0.078841 | 13.346 | 0.078454 | 0.078623 | ✓
462 | 1 | 472 |
| 26 | Accept | 0.080773 | 14.441 | 0.078454 | 0.079486 | ✓
488 | 1 | 976 |
| 27 | Accept | 0.082319 | 13.375 | 0.078454 | 0.079509 | ✓
467 | 3 | 351 |
| 28 | Accept | 0.081546 | 13.445 | 0.078454 | 0.079393 | ✓
469 | 1 | 251 |
| 29 | Accept | 0.07942 | 13.583 | 0.078454 | 0.079382 | ✓
469 | 1 | 583 |
| 30 | Accept | 0.079807 | 14.71 | 0.078454 | 0.079483 | ✓
493 | 1 | 1591 |

```

Optimization completed.

MaxObjectiveEvaluations of 30 reached.

Total function evaluations: 30

Total elapsed time: 256.1103 seconds

Total objective function evaluation time: 252.3732

Best observed feasible point:

NumLearningCycles	MinLeafSize	MaxNumSplits
492	1	1617

Observed objective function value = 0.078454
Estimated objective function value = 0.079483
Function evaluation time = 15.1003

Best estimated feasible point (according to models):

NumLearningCycles	MinLeafSize	MaxNumSplits
492	1	1617

Estimated objective function value = 0.079483
Estimated function evaluation time = 15.2835

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